

Composition is a degree-two datum

Locating compositional invariants by cohomological degree — a program, with three proved instances on directed containers

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Abstract

A directed container is a small category, and a question about such a category can be a question about its underlying shape, about a resource transported across it, or about how its morphisms compose. We observe that three invariants we have recently settled — one for each of these — sit at three distinct cohomological altitudes, and that the altitude of an invariant is a faithful record of how much of composition it is able to see. Out-degree homogeneity is a degree-zero invariant and is provably blind to composition. Supply-chain inventory consistency is a degree-one class on a resource presheaf, provably independent of the degree-two weld class that governs whether a chain factors at all. Composition itself — the Zappa–Szép weld — is a degree-two datum, and the comultiplication that encodes it supports a strictly finer invariant that does see composition: the left-translation trivialization of the tangent bundle detects exactly the groupoids, and, one notch finer, the thin groupoids, stratifying the degree-two layer by a discrete holonomy class. We assemble these as a single ladder and propose, as a program rather than a theorem, that cohomological degree classifies compositional invariants by what they can detect.

1 Introduction

One functor, three questions. A directed container is, by the theorem of Ahman and Uustalu [2], exactly a small category: the comonoids for the composition product $\triangleleft$ on polynomial functors are small categories, with objects the shapes and, for each object a , the direction set E_a as the set of morphisms out of a . A single such category carries structure at several levels at once. There is its *underlying functor* $p = \sum_a y^{E_a}$, which remembers only the object-indexed family of direction sets. There is *data transported across it* — a presheaf of resources, say stock levels at the nodes of a supply chain. And there is its *composition* — the comultiplication $\delta: p \rightarrow p \triangleleft p$, which records how two morphisms compose, and, for a composite built as a Zappa–Szép product, the weld that glues two factors together.

Questions about a category can interrogate any of these levels, and a recurring experience of this programme has been that an invariant designed to answer one kind of question turns out to

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be *structurally unable* to answer another. The derivative of a container is blind to its composition. The consistency of a transported resource is independent of whether the composite even factors as claimed. These are not accidents of proof technique; they are facts about where the relevant datum lives.

The observation. This note makes that experience precise by a single organizing device: grade a compositional invariant by the *cohomological degree* of the datum it reads.

- **Degree 0** is the underlying functor $U(p)$ — the family of out-degrees $|E_a|$, read by the Abbott–Altenkirch–Ghani–McBride derivative ∂ and by anything that factors through the forgetful $U: \mathbf{Cat} \rightarrow \mathbf{Poly}$. No composition data is present.
- **Degree 1** is a class $[\theta_R] \in H^1(\mathcal{S}; R)$ of a resource presheaf R over the nerve of the category — a route/reconvergence discrepancy.
- **Degree 2** is composition itself: the comultiplication δ , and the Zappa–Szép weld obstruction $[\omega] \in H^2(\mathbf{Sk}_\mathcal{S}; \mathcal{D})$ measuring whether a composite closes into the claimed factorization.

We then observe that three invariants we have proved, independently and for their own sakes, pin three points of this ladder, and that two sharp *non-collapse* results — degree-zero blindness, and the logical independence of the degree-one and degree-two classes — are exactly what give the grading teeth. The slogan is:

The cohomological degree at which a compositional invariant lives records what it can and cannot see. Composition — the Zappa–Szép weld — is a degree-two datum, and an invariant living below degree two is provably blind to it.

This is a program, not a theorem. We want to be scrupulous about what is and is not claimed. The grading above is a *heuristic* — an organizing principle that fits the proved instances and sharpens the grant’s theory narrative. The general statement, that *every* compositional invariant is classified up to detection power by its degree, is **open**; we state it in §7 as a conjecture and research direction, never as a result. What is proved is the three instances and the two non-collapse facts, each in a companion note or proof file cited below and *not* re-derived here. This is a positioning note: its job is to make the pattern visible, not to establish new mathematics.

The three proved pins, and a capstone. In the language of the grading:

- **Degree 0 (§3).** “Tangent-regularity” — all tangent fibres of Tp isomorphic — is equivalent to out-degree homogeneity, factors through $U: \mathbf{Cat} \rightarrow \mathbf{Poly}$, and is therefore *provably blind* to composition: the idempotent monoid $\{1, e\}$ and the group $\mathbb{Z}/2$ have the same underlying functor y^2 and are indistinguishable to it, though only the latter is a groupoid [5].
- **Degree 1 (§4).** Supply-chain inventory consistency is the vanishing of a degree-one class $[\theta_R] \in H^1(\mathcal{S}; R)$ on the resource presheaf, and it is *logically independent* of the degree-two weld class $[\omega] \in H^2(\mathbf{Sk}_\mathcal{S}; \mathcal{D})$: each can vanish while the other does not [6].
- **Degree 2 (§5).** The weld $[\omega]$ is composition’s own degree-two class. And the comultiplication supports an invariant that *does* see composition: the left-translation trivialization of the (vertical layer of the) tangent bundle exists over a component exactly when that component is a *groupoid* [7]. This separates $\mathbb{Z}/2$ from $\{1, e\}$ — the very pair degree zero cannot separate.

- **Capstone (§6).** One notch finer, the same trivialization lands on a *constant* presheaf exactly when the component is a *thin* groupoid, so the composition layer carries its own internal stratification — generic $\subsetneq$ groupoid $\subsetneq$ thin groupoid — separated by the vertex group $\text{Aut}(a)$ as a discrete holonomy class [8].

Method, and the one honest subtlety. The through-line is a single slogan about two structure maps of the same functor p . The derivative ∂ reads the *directions* E_a and nothing else — degree-zero shape. The comultiplication $\delta: p \rightarrow p \triangleleft p$ reads the *composition* — it lands in the two-fold composite $p \triangleleft p$, which is exactly why composition is a degree-two datum. The same container, viewed through ∂ or through δ , sits at two different cohomological altitudes, and the proved instances measure the gap between them.

The subtlety to flag at once: **the degrees do not collapse into one another.** There is no suspension isomorphism silently relating a degree-one presheaf class to a degree-two weld class; the simplest witness is the free loop $B\mathbb{Z}$, where $H^1(B\mathbb{Z}; A) = A$ but $H^2(B\mathbb{Z}; A) = 0$. The content of the degree-one/degree-two instance (§4) is precisely that these two classes vary independently, and the reader should resist the temptation — which the suggestive word “ladder” invites — to read the grading as a filtration of a single theory. It is a classification of *data*, not of a spectral sequence.

Outline. Section 2 fixes containers as categories, the two structure maps ∂ and δ , the resource presheaf and its H^1 , the weld and its H^2 , and the degree grading. Sections 3–5 walk the three proved pins, degree by degree, stating each result and the discriminating example that carries its intuition and citing the companion note for the proof. Section 6 adds the capstone monodromy tower. Section 7 states the program as a program, the open classification conjecture, and the sharpened applications story.

2 Preliminaries

We work with polynomial functors in one variable and finite support; everything is first order.

Containers as categories. A *container* in one variable is a polynomial functor $p = \sum_{a \in \text{Ob}} y^{E_a}$ with a set Ob of *shapes* and, for each shape a , a set E_a of *directions*; its extension sends a set X to $\sum_a X^{E_a}$ [1, 3]. The composition product $\triangleleft$ has unit y and realizes substitution. By Ahman and Uustalu [2] the comonoids in $(\text{Poly}, \triangleleft, y)$ are exactly small categories: a comonoid structure $\delta: p \rightarrow p \triangleleft p$ chooses identities (the counit) and a composition (the comultiplication), Ob is the object set, and $E_a = \coprod_c \text{Hom}(a, c)$ is the set of morphisms out of a , so $|E_a|$ is the *out-degree* of a . Write $\text{Comon}(\text{Poly}, \triangleleft) \simeq \text{Cat}$ for this equivalence and $U: \text{Cat} \rightarrow \text{Poly}$ for the forgetful functor that discards δ and keeps only the underlying functor $\sum_a y^{E_a}$. We compose *diagrammatically*: for $g: a \rightarrow b$ and $h: b \rightarrow c$, the composite $g \cdot h: a \rightarrow c$ means “ g then h ”.

Two structure maps: ∂ reads shape, δ reads composition. The one-hole derivative [1] is

$$\partial p = \sum_a \sum_{e \in E_a} y^{E_a \setminus \{e\}} = \sum_a |E_a| \cdot y^{|E_a|-1}.$$

It depends only on the direction sets, i.e. only on $U(p)$: it is a construction on the *underlying functor*. With $W = \mathbb{Z}[\varepsilon]/(\varepsilon^2)$ the dual-number Weil algebra, the attached tangent functor is the

Weil base change $T = (-) \otimes W$, computed on the first-order layer by substituting the infinitesimal and truncating,

$$Tp = p(y + \varepsilon v) \bmod \varepsilon^2 = p + \varepsilon v \cdot \partial p. \quad (1)$$

We write $(-) \otimes W$ rather than “ $(-) \triangleleft D$ ”: substituting $D = y + \varepsilon v$ and truncating is only the first-order shadow of the base change, and the two descriptions agree on the layer we use [11, 5]. By contrast the comultiplication $\delta: p \rightarrow p \triangleleft p$ is *not* a construction on $U(p)$: it is extra structure, and it lands in the two-fold composite $p \triangleleft p$. This is the precise sense in which composition is a degree-two datum — it is the structure map valued in the double composite — and it is the datum that ∂ cannot see.

The resource presheaf and its degree-one class. Model a supply chain as a small category $\mathcal{S}$: objects are nodes, morphisms out of a are admissible operations, composition is sequencing. *Inventory* is data transported across $\mathcal{S}$: a presheaf $R: \mathcal{S}^{\text{op}} \rightarrow \mathbf{Ab}$ of stock-groups, with $R(f)$ the transport along an operation f . A *global section* is a family $(x_a \in R(a))_a$ with $R(f)(x_{\text{cod } f}) = x_{\text{dom } f}$ for every f ; for a transport local system, where each $R(f)$ is translation by δ_f , a global section is a node-potential φ with $\delta_f = \varphi(\text{cod } f) - \varphi(\text{dom } f)$, i.e. the 1-cochain δ is a coboundary. The obstruction is its class

$$[\theta_R] := [\delta] \in H^1(\mathcal{S}; R).$$

This is a degree-one class, with coefficient system the resource R , measuring route discrepancy over the nerve of $\mathcal{S}$.

The weld and its degree-two class. A composite chain is a *Zappa–Szép product* $\mathcal{S} = \mathcal{C} \bowtie \mathcal{D}$ (procurement $\bowtie$ logistics): $\mathcal{C}, \mathcal{D}$ are wide subcategories, meeting in the discrete category $\mathcal{O}$ on the shared objects, and every morphism factors uniquely as $f = c \cdot d$ [2, 9]. The matched-pair law $\lambda: \mathcal{D}\mathcal{C} \Rightarrow \mathcal{C}\mathcal{D}$ is the distributive rewriting. Whether the transversal closes into such a factorization is governed by the *weld obstruction*

$$[\omega] \in H^2(\text{Sk}_{\mathcal{S}}; \mathcal{D}),$$

a Baues–Wirsching 2-cocycle on the orbit category with coefficients the vertex-group presheaf $\mathcal{D}$; the chain decomposes as claimed exactly when $[\omega] = 0$ [9]. This is a degree-two class, with coefficient system $\mathcal{D}$ (the invertible/weld part), depending only on the category and the chosen factorization — no resource data enters. It is the home, for instance, of the blockchain re-entrancy obstruction $[\omega] = \varepsilon \in H^2 \cong \mathbb{Z}/2$ [10].

The degree grading. We rank a compositional invariant by the degree of the datum it reads: degree 0 for the underlying functor $U(p)$ (equivalently the out-degree family), degree 1 for a class in $H^1(\mathcal{S}; R)$ of a resource presheaf, degree 2 for the comultiplication δ and the weld $[\omega] \in H^2$. We stress again (§1) that this is a grading of *data*; the degrees are genuinely different cohomological theories with different coefficient systems, and no dimension-shift relates them.

3 Degree zero: the invariant blind to composition

The lowest rung reads only the underlying functor, and is for that reason unable to see composition at all. The result is proved in full in the companion note [5]; we state it and exhibit the discriminating examples.

The tangent bundle (1) has, over a point of shape a , a fibre that is the free module on the direction set E_a , of rank $|E_a|$. Call p *tangent-regular* if these fibres are all isomorphic to a single

module — the honest residue, in the representable/monoidal setting, of Lanfranchi’s parallelizability [4, 5].

Theorem 1 ([5]). *A container $p = \sum_a y^{E_a}$ is tangent-regular iff the out-degree $a \mapsto |E_a|$ is constant on Ob . This condition depends only on $U(p)$ — it factors through $U: \text{Cat} \rightarrow \text{Poly}$ — and is therefore blind to composition: no property of the comultiplication δ can be equivalent to it.*

Example 2 (the blindness, made concrete). The idempotent monoid $\{1, e\}$ with $e^2 = e$ and the group $\mathbb{Z}/2$ both have underlying functor y^2 . They have the *same* tangent bundle and are both tangent-regular (one object, so out-degree is trivially constant), yet $\mathbb{Z}/2$ is a groupoid and $\{1, e\}$ is not. Degree zero cannot tell them apart. Conversely the groupoid $\mathbb{Z}/2 \sqcup \mathbb{Z}/3$ has out-degrees 2 and 3, so $U = y^2 + y^3$ is *not* tangent-regular though it is a groupoid: being a groupoid neither implies nor is implied by tangent-regularity.

The cautionary form of this rung is easy to fall for. A genuine theorem of Lanfranchi [4] — an object of a *Cartesian* tangent category is parallelizable iff it is a Lie group object — invites the port “a container is parallelizable iff it is a groupoid.” The port is ill-posed, because the container derivative’s tangent structure is the monoidal base change $(-) \otimes W$ and not a Cartesian one [5]; what survives of it is tangent-regularity, which lives at degree zero and cannot see the groupoid property. This is no defect in [4] — that theorem is about Cartesian tangent categories — but exactly the symptom of asking a shape-level invariant a composition-level question. Connected groupoids *are* tangent-regular, the faithful shadow of “Lie group $\Rightarrow$ parallelizable”; but strictly more categories are, and the gap is the composition content that only degree two recovers (§5).

4 Degree one: a class independent of the weld

The middle rung reads a resource transported across the category. Its invariant is genuinely cohomological — a class in H^1 — and the central fact is that it is *independent* of the degree-two weld class, both as a matter of logic and, more tellingly, because the two live in different degrees over different coefficient systems. The results are proved in [6]; we state them.

Theorem 3 ([6]). *Let $\mathcal{S} = \mathcal{C} \bowtie \mathcal{D}$ and let R be a resource local system of abelian stock-groups. Global inventory is consistent — R admits a global section over the sub-chain cover $\{\mathcal{C}, \mathcal{D}\}$ — iff $[\theta_R] = 0 \in H^1(\mathcal{S}; R)$. The matched-pair law λ plays no role: a family compatible with the $\mathcal{C}$ -morphisms and the $\mathcal{D}$ -morphisms is a global section, because compatibility is closed under composition and $\mathcal{C} \cup \mathcal{D}$ generates $\mathcal{S}$.*

Theorem 4 (separation, [6]). *$[\theta_R] \in H^1(\mathcal{S}; R)$ and $[\omega] \in H^2(\text{Sk}_{\mathcal{S}}; \mathcal{D})$ are logically independent: both “ $[\omega] = 0$ with $[\theta_R] \neq 0$ ” and “ $[\omega] \neq 0$ with $[\theta_R] = 0$ ” occur. Hence inventory consistency is not the vanishing of the weld class, in either direction.*

Example 5 (the two witnesses). For the first: let $\mathcal{S}$ be the free category on the diamond quiver $a \rightrightarrows \{b, c\} \rightrightarrows d$ with distinct routes. It is directed-acyclic, so it has no non-identity invertibles, the weld coefficient system is trivial, and $[\omega] = 0$ is forced; yet $B\mathcal{S}$ has first Betti number 1, so a transport system with nonzero loop holonomy gives $[\theta_R] \neq 0$ — the chain decomposes trivially but stock fails to glue across the two routes. For the second: the rigid-twist category with $\text{End}(a) = \mathbb{Z}/2$ has $[\omega]$ the generator of $H^2(\text{Sk}; \mathbb{Z}/2)$, nonzero, while the zero inventory system is consistent, $[\theta_R] = 0$. The two classes vary independently; they cannot even be compared by a natural map, since in the first witness the weld coefficient $\mathcal{D}$ is trivial while the resource coefficient R is not.

Remark 6 (why there is no bridge). It is worth seeing why Theorem 4 is not an artifact. A rescue of “consistent iff $[\omega] = 0$ ” would need a natural isomorphism $H^1(\mathcal{S}; R) \cong H^2(\text{Sk}_{\mathcal{S}}; \mathcal{D})$ identifying the two classes. No such dimension shift exists in general: for the free loop $\mathcal{S} = B\mathbb{Z}$ one has $H^1(B\mathbb{Z}; A) = A$ while $H^2(B\mathbb{Z}; A) = 0$, so the degrees genuinely do not match. This is the same “rhyme, not identity” caution that governs the two H^2 -sites elsewhere in the programme [9], here one degree down: the same cyclic group can appear on both rungs, but the site, the coefficients, and now the *degree* differ, and one must check all three before identifying two obstructions.

The conjecture tested in [6] — that inventory consistency *equals* the weld $[\omega] = 0$ — was a false generalization of blockchain re-entrancy, where the conserved quantity *is* the weld-state itself and so no separate resource presheaf exists. In a supply chain the stock lives on the nodes, transported by operations, and is a genuinely separate degree-one datum. The modelling is not debunked but sharpened: re-entrancy is a degree-two failure (handoff order, in the weld), supply-chain inventory inconsistency is a degree-one failure (route discrepancy, on the nodes), and they are different economic failure modes at different altitudes.

5 Degree two: composition, and the invariant that sees it

The top rung is composition itself. It appears in two guises. First, the weld $[\omega] \in H^2$ of §2 *is* a degree-two class, measuring whether composition closes into the asserted factorization; re-entrancy [10] is its economic witness. Second — and this is what promotes “composition is a degree-two datum” from a slogan about the weld to a statement about every container — the comultiplication δ supports a trivialization of the tangent bundle that *does* see composition, exactly where the degree-zero invariant of §3 was blind. The result is proved in [7].

Fix a morphism $g: a \rightarrow b$ and pin the first slot of δ to it: this is the δ -translation

$$L_g: E_b \rightarrow E_a, \quad L_g(h) = g \cdot h,$$

precomposition by g . It is contravariantly functorial, assembling the comultiplication into the *arrows-out presheaf* $E_{(-)}: \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$, $a \mapsto E_a$, $g \mapsto L_g$ — a property-free restatement of δ . (The statements below concern this vertical/linear layer of the tangent object $Tp = p \otimes W$, not the full bundle [7].) Say Tp is δ -trivializable over a connected component $[a]$ if $E_{(-)}$ restricted there is a *local system* — every L_g a bijection — the honest discrete analogue of trivializing a tangent bundle by translations, $TG \cong G \times \mathfrak{g}$.

Theorem 7 ([7]). *Tp is δ -trivializable over $[a]$ iff the full subcategory on $[a]$ is a groupoid; globally, iff $\mathcal{C}$ is a groupoid. Surjectivity of the translations already forces full invertibility — a surjective L_g hits the identity $\text{id}_a \in E_a$, and a morphism composing to an identity is an inverse — so the dividing line is exactly “groupoid”, with no weaker cancellation condition in between.*

Example 8 (seeing what degree zero could not). Return to the pair of Example 2. On $\mathbb{Z}/2$ every L_g is a bijection, so Tp is δ -trivializable. On $\{1, e\}$ with $e^2 = e$ the translation L_e sends $1 \mapsto e$ and $e \mapsto e$, missing $\text{id} = 1$; it is not even injective, and $\{1, e\}$ is not δ -trivializable. The degree-two invariant *separates* $\mathbb{Z}/2$ from $\{1, e\}$ — the exact pair that the degree-zero invariant, reading only the common underlying functor y^2 , was structurally unable to tell apart (Theorem 1). This is the content of “composition is a degree-two datum” as a theorem rather than a slogan: there is a genuine invariant, built from δ , that detects a composition property, and it provably could not have been built at degree zero.

The reason the degree-two invariant succeeds where degree zero fails is now transparent: L_g is built from δ , the structure map valued in $p \triangleleft p$, whereas tangent-regularity was built from ∂ , a construction on $U(p)$. “Groupoid”, “connected”, “invertible” are properties of δ ; only an invariant that reads δ can see them, and reading δ is reading degree two.

6 The capstone: a monodromy tower inside degree two

Degree two is not a single stratum. The δ -translations can be asked to do more than be invertible: they can be asked to trivialize to a *constant* identification, with trivial monodromy. This finer demand detects a strictly smaller class of categories, and the obstruction between the two is a discrete holonomy class. The result is proved in [8].

Call $E_{(-)}$ *constant-trivializable* over $[a]$ if it is naturally isomorphic to a *constant* presheaf Δ_V : bijections $\varphi_b: E_b \rightarrow V$ with $\varphi_x \circ L_g = \varphi_y$ for every $g: x \rightarrow y$ of the component, loops included. Read against a loop, this condition is monodromy: it asks the translation cocycle $\{L_g\}$ to be a coboundary.

Theorem 9 ([8]). *$E_{(-)}$ is constant-trivializable over $[a]$ iff $[a]$ is a thin (indiscrete) groupoid — a groupoid with trivial vertex groups, equivalently an equivalence-relation component. Globally, iff $\mathcal{C}$ is a thin groupoid. Constant-trivializability implies δ -trivializability, so the groupoid hypothesis of Theorem 7 comes for free; applying naturality to a loop $g \in \text{Aut}(a)$ forces $L_g = \text{id}$ and hence, because the regular action fixes the basepoint $\text{id}_a \in E_a$, forces $g = \text{id}_a$ — the vertex group vanishes.*

Corollary 10 (the three-level monodromy tower). *Over a connected component, three successively finer conditions on the vertical layer $E_{(-)}$ stratify the composition layer:*

Rung	Condition on $E_{(-)} _{[a]}$	Equivalent to	Status
1	generic presheaf (each L_g a function)	nothing	trivial
2	local system (every L_g a bijection)	$[a]$ a groupoid	[7]
3	$\cong$ constant presheaf	$[a]$ a thin groupoid	[8]

Rungs 2 and 3 are distinct, separated by a nontrivial vertex group: $\mathbb{Z}/2$ (one object, $\text{Aut} = \mathbb{Z}/2$) is a groupoid but not constant-trivializable, since the regular representation $\mathbb{Z}/2 \curvearrowright \mathbb{Z}/2$ is free and the loop monodromy is nontrivial; the indiscrete category I_2 (two objects, trivial vertex group) is both. A two-object groupoid with vertex group $\mathbb{Z}/2$ is rung 2 but not rung 3, confirming that the obstruction is the vertex group, not one-object-ness.

So the composition datum carries an internal grading of its own. Rung 2 is the honest discrete analogue of parallelizability — trivialize by translations, as every Lie group admits, including those with nontrivial π_1 . Rung 3 is strictly stronger: parallelizable *with a global flat framing*, trivial holonomy, which in the discrete setting forces the vertex group — the discrete holonomy group — to vanish. The obstruction to upgrading rung 2 to rung 3 is exactly $\text{Aut}(a)$ as a monodromy class. Thus “composition is a degree-two datum” acquires an internal stratification: shape (degree zero) $<$ groupoid $<$ thin groupoid, with a single nontrivial vertex group marking the step from the second to the third. We are careful not to overclaim: rungs 2 and 3 are composition-*sensitive* invariants of δ , and $\text{Aut}(a)$ is a genuine holonomy class, but these are properties of the comultiplication, not themselves identified with a literal H^2 cohomology class. The literal cohomological anchors of the ladder remain $[\theta_R] \in H^1$ and $[\omega] \in H^2$; the tower refines the *datum* δ that lives at that altitude.

7 The program, and what it would take to prove

What is assembled here. Three invariants, each proved for its own sake, fall on a single ladder: out-degree homogeneity at degree 0, blind to composition; inventory consistency at degree 1, independent of the weld; and at degree 2 both the weld $[\omega]$ and the δ -translation invariant that — with its monodromy tower generic $\subsetneq$ groupoid $\subsetneq$ thin groupoid — sees exactly what degree zero could not, separating $\mathbb{Z}/2$ from $\{1, e\}$. The two non-collapse results — degree-zero blindness (Theorem 1) and $H^1 \perp H^2$ independence (Theorem 4) — are what make the grading informative rather than decorative: they show the rungs are genuinely different.

The program, stated as a program.

The degree program. For a directed container (small category) $\mathcal{C}$ and a compositional invariant I , the cohomological degree of the datum I reads determines what I can detect: an invariant that factors through the degree- d truncation of $\mathcal{C}$ ’s data is blind to every composition phenomenon that is genuinely of degree $> d$. In particular composition — the comultiplication δ and its weld $[\omega] \in H^2$ — is a degree-two datum, and no invariant below degree two can detect it.

This is a **conjecture and research direction, not a theorem**. We have three instances and two non-collapse facts; we do not have a definition of “the degree- d truncation of a category’s data” general enough to make the statement precise and uniform, nor a proof that detection power is monotone in degree in general. Making the program a theorem would require: (i) a functorial notion of the degree- d data of a directed container — plausibly the d -truncation of its nerve together with a coefficient system — such that $U: \text{Cat} \rightarrow \text{Poly}$ is the degree-0 case; (ii) a general “blindness” theorem that an invariant factoring through degree $< d$ cannot separate two categories agreeing to degree $d - 1$; and (iii) a converse realizing every genuine degree- d distinction by some invariant. Each of (i)–(iii) is open. We flag explicitly that the degrees do not collapse (Remark 6), so any such theorem must classify data, not assemble a single filtration.

The sharpened applications story. As a grant-narrative frame the ladder is immediately useful, and it *sharpens* rather than replaces the earlier picture. Compositional failure is not one phenomenon but several, cleanly separated by where the conserved quantity lives:

Datum lives...	Failure class	Economic instance
on the shape (out-degrees)	degree 0 — invisible to ∂	tangent/shape geometry
on the nodes (transported)	$[\theta_R] \in H^1$ — route discrepancy	supply-chain inventory
in the weld (composite’s state)	$[\omega] \in H^2$ — handoff order	blockchain re-entrancy

Supply chains are thereby not a second instance of “failure = nonzero H^2 ” but the first clean instance of “failure = nonzero H^1 ”; re-entrancy remains the H^2 witness; and the tangent/shape invariant marks the degree-zero floor below which composition is simply not recorded. Two economic failure modes, cleanly separated by degree, each with a witness — and a precise account of which invariants can and cannot detect which.

Acknowledgement and honesty. The port of Lanfranchi’s parallelizability [4] and the supply-chain conjecture of [6] are both *sharpened* here, not debunked: the first because its home is a Cartesian tangent category and the container derivative’s is not, the second because inventory is a node-level degree-one datum and not the weld. The degree program itself is a heuristic that

organizes proved facts; its general form is open. We have taken care that every degree assignment in this note tracks a datum we have actually computed, and that the slogan “composition is a degree-two datum” is underwritten, not merely asserted, by Theorems 7 and 9.

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